

Marwan Abudaif

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SUMMARY

Computer Engineering senior (GPA 3.48/4.0, AUC) shipping production software at METI Software — sole engineer on a multi-tenant Laravel + Next.js client portal, and owner of the chat, dataset, and settings surfaces of a geospatial AI platform. Outside work: 45k lines of real-time 3D engine code with WebRTC multiplayer, a native iOS app, and a senior thesis measuring LLM sycophancy.

EDUCATION

B.S. in Computer Engineering

The American University in Cairo (AUC)

Expected Dec 2026

Cairo, Egypt

- GPA 3.48/4.0 • Coursework: Algorithms, Operating Systems, Computer Architecture, Databases, Machine Learning, Distributed Systems • Study abroad: Universidad Carlos III de Madrid (Spring 2025).

SKILLS

- **Languages:** TypeScript / JavaScript, Python, PHP, C++, Swift, SQL, Kotlin, HTML5 / CSS3
- **Frontend:** React 19, Next.js 16, Angular 22, Svelte 5, Tailwind v4, Three.js, shadcn / Base UI
- **Backend & Data:** Laravel 12, Node.js, FastAPI, PostgreSQL + PostGIS, MySQL, Firebase, REST, SSE, WebSockets, WebRTC
- **Infra, Quality & AI:** Docker, AWS, Bitbucket Pipelines CI, Playwright, Vitest, PHPUnit, Git • LLM evaluation & harness design, RAG, MCP agent tooling, Claude & Gemini APIs
- **Certification:** AZ-900 Azure Fundamentals • **Work authorization:** Egypt + UAE residency

EXPERIENCE

Software Engineer

METI Software (Micro Engineering Technologies)

Jun 2025 – Present

Remote – Canada

- Built core surfaces of the **AI Author Tool** (Angular), which parses uploaded documents into structured courses — the Lesson → Sections → Bricks hierarchy, drag-and-drop reordering, content bricks, **English/Arabic localisation with RTL**, and an SSE course-generation progress stream.
- On **Aqtar**, a Saudi-facing **geospatial services marketplace**, shipped the Ratings & Reviews module and the content reporting/moderation system end-to-end (**195 commits**, ten-engineer team), plus the English/Arabic localisation layer.
- Promoted from Frontend Intern; now **sole engineer** on the **MVS Client Portal**, a multi-tenant Laravel 12 + Next.js 16 product spanning nine features and three roles — 30k lines of PHP, 35k of TS, 30 migrations, 49 test classes.
- Designed an **engagement-health scoring service** computed on read rather than stored, collapsing a naive 21-query-per-screen implementation into 3 queries across a 6-month trend.
- Enforced a role-scoped authorization matrix (404-not-403 so record IDs cannot be probed) and swept **every route against every role** in Playwright to prove the matrix was not vacuous.
- On **GuloGulo**, a geospatial AI platform (Next.js 16 / React 19 / Laravel / PostGIS), owned chat, datasets, and settings — 45 commits and **8 merged pull requests** — including a resumable **multipart upload engine** for 40 GiB archives (batched part signing, bounded concurrency, per-part retry) behind an SSE transport that degrades to polling under full-jitter backoff.

Software Engineering Intern

Pulse (formerly Giza Systems) — BOOKSHELF platform

Aug 2026

Cairo, Egypt

- Sole author of the delivered Angular 22 application (~3.8k lines): customer catalogue and admin workspace with signal-based state, typed reactive forms, and role guards over a six-route access matrix.
- Wrote the spec the team built against (versioned REST contract, consistent error envelope, security decisions) and met the accessibility and 390/1024/1440px responsive requirements.

Fixed Networks R&D Intern

Nokia & WE Telecom — exploratory placement on FTTH predictive diagnostics (Python, simulator data)

Aug 2025

Cairo, Egypt

FEATURED PROJECTS

GEM Tours (Project Horus) — AI Guide for the Grand Egyptian Museum

2026

Next.js 16 • Vertex AI • Firestore vector search • Gemini • Capacitor

- Built solo (18k lines, 72 TypeScript modules): point a phone at an artifact and hear its history in your language. Vertex AI multimodal embeddings in a **Firestore vector index** identify it in **under 300 ms**, detection in a web worker so the preview never stalls, with EN/AR translation and RTL at scan time.
- Treated inference cost as an architectural constraint: a three-tier cache ending in a **shared Firestore store**, so **one visitor's generation serves every later visitor**, plus an on-device TTS fallback — ~\$0.008 per visitor.

GeoFighters 2.0 — Browser-Native Co-op 3D Shooter

[GitHub](#)

TypeScript • Three.js • Svelte 5 • Rapier • miniplex ECS • WebRTC

- 45k lines: ECS engine with instanced rendering, Rapier physics, PWA and Capacitor iOS builds. Stabilised frame time via instancing, spatial-hash separation and a pre-work frame limiter (160 → 59 fps).
- Designed a host-star **WebRTC** topology — 30 Hz snapshots on unreliable/unordered SCTP channels so a dropped packet never stalls the next, per-peer fallback to a socket relay when ICE fails, and TURN on 443/TCP after finding CGNAT is the norm on Egyptian and UAE carriers.

TELOS II — Senior Thesis: Measuring LLM Sycophancy

Aug 2026 – Present

FastAPI • Python • React • TypeScript • Leading a team of 7

- Architected a system where **the model renders but never decides** — action authority sits in a versioned, SHA-256-pinned policy behind an independent guard.